AI Agent Architecture: Prompt and Response

User prompt

does an agent usually consist of a software program making LLM requests in a programmed loop

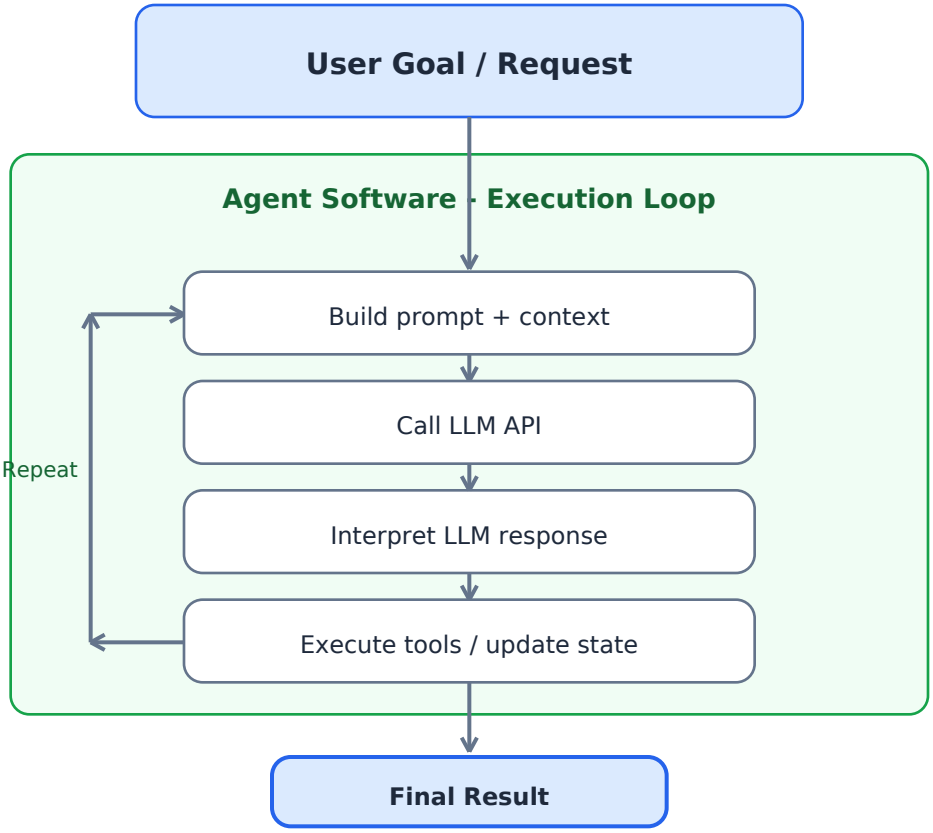
Assistant response

Yes. That is a good description of how many LLM-based AI agents are implemented.

An AI agent typically consists of a software program that repeatedly calls an LLM, evaluates its responses, executes requested actions, and continues until a goal is achieved or a stopping condition is reached.

The important distinction is that the LLM provides reasoning and decision-making, while the surrounding software controls execution.

Typical AI Agent Architecture



Simplified Agent Implementation

In Python-like pseudocode:

```
def run_agent(goal):
    context = [goal]
```

```
while True:
    # Ask LLM what to do next
    response = call_llm(context)

    # Has the agent finished?
    if response.is_complete:
        return response.result

    # Execute the requested action
    result = execute_tool(
        response.tool,
        response.arguments
    )

    # Feed result back to LLM
    context.append(result)
```

The software loop is responsible for managing state, invoking tools, enforcing limits, and deciding when to terminate. The LLM is responsible for proposing the next action or generating the final answer.

Important architectural distinctions

Component	Responsibility
Agent program	Orchestrates the execution loop
LLM	Interprets context, reasons, selects actions
Tool interface	Provides access to external capabilities
Agent state	Maintains conversation and intermediate results
Termination logic	Determines when execution stops

There are variations. Some agents use a fixed workflow rather than an open-ended loop. Others employ multiple cooperating agents or use LLM APIs that manage portions of the tool-calling sequence.

The essential architectural idea is that the agent is the entire executing system, not just the LLM. The LLM is one component that the agent software calls as needed.

This distinction is particularly useful when explaining the difference between a chatbot and an AI agent in a software architecture presentation.